

# Class 10 Mathematics - Key Concepts

## Chapter 1: Real Numbers

### Euclid's Division Lemma

For any two positive integers  $a$  and  $b$ , there exist unique integers  $q$  and  $r$  such that:

$$a = bq + r, \text{ where } 0 \leq r < b$$

This is the basis for finding HCF (Highest Common Factor) of two numbers.

### Fundamental Theorem of Arithmetic

Every composite number can be expressed as a product of primes, and this factorization is unique.

Example:  $84 = 2 \times 2 \times 3 \times 7 = 2^2 \times 3 \times 7$

## Chapter 2: Polynomials

### Polynomial Definition

A polynomial  $p(x)$  in one variable  $x$  is an algebraic expression of the form:

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0$$

Where  $a_n, a_{n-1}, \dots, a_0$  are real numbers and  $n$  is a whole number.

### Zeros of a Polynomial

A zero of a polynomial  $p(x)$  is a number  $\alpha$  such that  $p(\alpha) = 0$ .

## Chapter 3: Linear Equations in Two Variables

Standard form:  $ax + by + c = 0$

Where  $a, b, c$  are real numbers and both  $a$  and  $b$  are not zero.

Solution: Any pair  $(x, y)$  that satisfies the equation.

## Chapter 4: Quadratic Equations

Standard form:  $ax^2 + bx + c = 0$ , where  $a \neq 0$

### Quadratic Formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Discriminant:  $D = b^2 - 4ac$

- If  $D > 0$ : Two distinct real roots
- If  $D = 0$ : Two equal real roots
- If  $D < 0$ : No real roots

## Chapter 5: Arithmetic Progressions (AP)

An arithmetic progression is a sequence where each term after the first is obtained by adding a constant difference.

General term:  $a_n = a + (n-1)d$

Where  $a$  = first term,  $d$  = common difference,  $n$  = term number

Sum of first  $n$  terms:  $S_n = n/2[2a + (n-1)d]$

## Chapter 6: Triangles

### Pythagoras Theorem

In a right-angled triangle, the square of the hypotenuse equals the sum of squares of the other two sides.

$$c^2 = a^2 + b^2$$

### Similar Triangles

Two triangles are similar if:

1. Their corresponding angles are equal
2. Their corresponding sides are proportional

Criteria for similarity:

- AAA (Angle-Angle-Angle)
- SSS (Side-Side-Side)
- SAS (Side-Angle-Side)

### Practice Problems

1. Find the HCF of 96 and 404 using Euclid's algorithm.
2. Find the zeros of polynomial:  $p(x) = x^2 - 5x + 6$
3. Solve quadratic equation:  $2x^2 - 7x + 3 = 0$
4. Find the 10th term of AP: 2, 7, 12, 17, ...

### Important Formulas

- Area of triangle =  $1/2 \times \text{base} \times \text{height}$
- Heron's formula:  $\sqrt{s(s-a)(s-b)(s-c)}$ , where  $s = (a+b+c)/2$
- Distance formula:  $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Section formula:  $((mx_2 + nx_1)/(m+n), (my_2 + ny_1)/(m+n))$

Remember: Practice regularly and understand the concepts rather than memorizing!